

Complex Engineering Problem

Course Code and Title: EL3032 Power Electronics

Semester: Spring 2025

Instructor: Dr. Muhammad Naeem

Total Marks:

Deadline: May 05, 2025

CLOs and PLOs for Complex Engineering Problem

CLOs		Description	Taxonomy Levels	PLO
CLO 4	Theory	Design Power Converter Circuits	C5	3

Instructions

Copied Solution or late submission will get no credit. Use of any AI-based platform to generate or assist with any part of the problem solution, including design decisions, calculations, or report writing is strictly prohibited.

This problem is designed to assess individual understanding, analytical ability, and engineering design skills. Using AI tools bypasses this learning process and compromises academic integrity.

Problem Statement

Design a single output DC-DC converter for a next generation wearable sensor node used in a health monitoring application. The device is powered by a single-cell Li-ion battery (3.7 nominal, 3.0 minimum, 4.2 maximum) and requires a regulated output of +5 V at 300 mA to power a sensor module and its wireless transmitter.

Objectives

1. Select a suitable DC-DC converter topology (buck-boost, Ćuk or flyback) based on application constraints.
2. **Design** a complete converter circuit that meets specified electrical, thermal and cost requirements.
3. Apply analytical skills to calculate and verify component values, operating parameters, and converter performance.
4. Analyze trade-offs in power converter design, including efficiency vs size and cost vs performance.

Deliverables

Technical Design Report

- Introduction and application context
- Justification of selected topology with comparison to alternatives
- Complete circuit schematic
- Design calculations
- Component selection with cost estimates
- Efficiency and thermal considerations

Design Requirements and Constraints

- Input voltage range: 3.0 V to 4.2 V
- Output voltage: +5 V $\pm 5\%$
- Output current: up to 300 mA
- Efficiency target: > 85% at nominal load
- Size constraint: must be compact enough to fit within a wearable device
- Thermal constraint: No heat sink allowed
- Component cost: Power stage components should not exceed 3000/- PKR

Complex Engineering Problem Attributes

WP1: Depth of knowledge WP2: Range of conflicting requirements WP3: Depth of analysis WP4: Familiarity of issues WP5: Extent of applicable codes WP6: Extent of stakeholders WP7: Interdependence	WP1: Depth of Knowledge -- Analysis of power demand requires knowledge of WK3 and WK4. Design requires knowledge of WK5 WP2: Range of conflicting requirements -- Requirements are conflicting. All conveniences/facilities are desired, but power consumption should be minimized. Requires knowledge of WK7. WP3: Depth of analysis -- No unique solution. Multiple designs possible. Analysis needed to select the most appropriate one.			
	Rubrics		CLOs	Marks
	Understands objectives, requirements, constraints, and standards clearly	WP1, WP2	CLO4	20
	Analyzes power demand considering all requirements and constraints	WP1, WP2	CLO4	10
	Considers multiple solutions and selects suitable one based on analysis	WP1, WP3	CLO4	20
	Develops a final design which satisfies all requirements, constraints, and standards	WP1, WP3	CLO4	40
	Duly considers cost and thermal aspects as well.	WP1, WP2, WP3	CLO4	10
	Report			

Knowledge Profiles

WK3 - Engineering Fundamentals: A systematic, theory-based formulation of engineering fundamentals required in an engineering discipline.

WK4 - Engineering Specialization: The knowledge of Engineering specialization that provides theoretical frameworks and bodies of knowledge for the accepted practice areas that are at the forefront in a discipline.

WK5 - Engineering Design: The *Design Thinking Knowledge* that supports engineering design in a practice area of an engineering discipline.

WK7 - Engineering in Society: A systematic, comprehension-based knowledge of the role of engineers in a society and the professional issues related to practicing engineering profession in a discipline: ethics and the professional responsibility of an engineer to public safety including the impact of an engineering activity i.e. economic, social, cultural, environmental and sustainability